

1. Preliminaries

Non-central distributions and quadratic forms

If independent $X_i \sim \mathcal{N}(\mu_i, 1)$, then

$$\sum_{i=1}^n X_i^2 \sim \chi_n^2(\lambda), \quad \lambda = \sum_i \mu_i^2,$$

with $\mathbb{E}X = n + \lambda$, $\text{Var}(X) = 2(n + 2\lambda)$, and

$$M_X(t) = (1 - 2t)^{-n/2} \exp\left\{\frac{\lambda t}{1 - 2t}\right\}, \quad t < 1/2.$$

Quadratic forms. If $\mathbf{y} \sim \mathcal{N}(\mu, \Sigma)$ with $\Sigma \succ 0$ and A, A_1, A_2 are symmetric, then

- $\mathbf{y}'A\mathbf{y} \sim \chi_m^2(\mu'A\mu)$ iff $A\Sigma$ is idempotent, with $m = \text{rank}(A\Sigma) = \text{rank}(A)$;
- $\mathbf{y}'A_1\mathbf{y} \perp \mathbf{y}'A_2\mathbf{y}$ iff $A_1\Sigma A_2 = 0$;
- if $A\Sigma B = 0$, then $A\mathbf{y} \perp \mathbf{y}'B\mathbf{y}$.

Non-central t and F . If $Y \sim \mathcal{N}(\delta, 1)$, $W \sim \chi_n^2$, and $Y \perp W$, then $Y/\sqrt{W/n} \sim t_n(\delta)$. If $Y \sim \chi_m^2(\theta)$, $W \sim \chi_n^2$, and $Y \perp W$, then $(Y/m)/(W/n) \sim F_{m,n}(\theta)$.

Cochran's theorem

Suppose $X \sim \mathcal{N}(I_n, I_n)$, $Q_i = X'A_iX$, $A_i = A'_i$, $\text{rank}(A_i) = r_i$, and $\sum_{i=1}^k Q_i = X'X$. The following are equivalent:

$$Q_i \sim \chi_{r_i}^2(\mu'A_i\mu) \ \forall i, \quad \sum_i r_i = n, \quad Q_i \perp Q_j \ \forall i \neq j.$$

2. Linear models

Consider the homoskedastic model

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \mathbb{E}(\boldsymbol{\epsilon} \mid X) = 0, \quad \text{Var}(\boldsymbol{\epsilon} \mid X) = \sigma^2 I.$$

Estimability

A linear function $c'\boldsymbol{\beta}$ is estimable iff any of the following equivalent conditions holds:

- every normal-equation solution $\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'\mathbf{y}$ gives an unbiased $c'\hat{\boldsymbol{\beta}}$;
- $c'\boldsymbol{\beta}$ is invariant across generalized inverses / normal-equation solutions;
- $c \in \mathcal{C}(X')$;
- $\exists a$ such that $\mathbb{E}(a'\mathbf{y}) = c'\boldsymbol{\beta}$ for all $\boldsymbol{\beta}$.

Constrained OLS and Aitken model

Under $C\boldsymbol{\beta} = d$,

$$\hat{\boldsymbol{\beta}}_c = \hat{\boldsymbol{\beta}} - (X'X)^{-1}C'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\boldsymbol{\beta}} - d), \quad \hat{\boldsymbol{\beta}} = (X'X)^{-1}X'\mathbf{y}.$$

If $\text{Var}(\boldsymbol{\epsilon}) = \sigma^2 V$ with $V \succ 0$, transform by $V^{-1/2}$ or use GLS.

Asymptotics: homoskedastic fixed-design OLS

Drop normality but keep fixed design and homoskedastic independent errors. Assume p is fixed, X has full column rank, and

$$Q_n = \frac{X'X}{n} \rightarrow Q \succ 0, \quad \max_{1 \leq i \leq n} \frac{\|x_i\|}{\sqrt{n}} \rightarrow 0, \quad \mathbb{E}\varepsilon_i = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2,$$

with a Lindeberg / finite-moment condition. Then

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X'X)^{-1}X'\boldsymbol{\epsilon}, \quad \sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = Q_n^{-1} \frac{X'\boldsymbol{\epsilon}}{\sqrt{n}}.$$

By multivariate Lindeberg–Feller CLT,

$$\frac{X'\boldsymbol{\epsilon}}{\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_i x_i \varepsilon_i \xrightarrow{d} \mathcal{N}(0, \sigma^2 Q), \quad \sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}(0, \sigma^2 Q^{-1}).$$

Consequently, for fixed full-row-rank C ,

$$\sqrt{n}(C\hat{\boldsymbol{\beta}} - C\boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}(0, \sigma^2 CQ^{-1}C'),$$

and under $H_0 : C\boldsymbol{\beta} = d$,

$$W_n = \frac{(C\hat{\boldsymbol{\beta}} - d)'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\boldsymbol{\beta}} - d)}{\hat{\sigma}^2} \xrightarrow{d} \chi_q^2.$$

The usual residual variance remains consistent:

$$\hat{\sigma}^2 = \frac{\boldsymbol{\epsilon}'(I - P_X)\boldsymbol{\epsilon}}{n - p}, \quad \frac{1}{n} \boldsymbol{\epsilon}'P_X\boldsymbol{\epsilon} = \left(\frac{X'\boldsymbol{\epsilon}}{n}\right)' Q_n^{-1} \left(\frac{X'\boldsymbol{\epsilon}}{n}\right) = o_p(1),$$

so $\hat{\sigma}^2 \xrightarrow{p} \sigma^2$. Thus t -statistics are asymptotically standard normal and Wald statistics are asymptotically χ_q^2 .

3. Inference for Gaussian linear models

Assume $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 I_n)$ and write $P_X = X(X'X)^{-1}X'$.

$$\hat{\sigma}^2 = \frac{\mathbf{y}'(I - P_X)\mathbf{y}}{n - \text{rank } X}, \quad C\hat{\boldsymbol{\beta}} = C(X'X)^{-1}X'\mathbf{y}.$$

t -test for $H_0 : c'\boldsymbol{\beta} = d$

$$t = \frac{c'\hat{\boldsymbol{\beta}} - d}{\hat{\sigma}\sqrt{c'(X'X)^{-1}c}} \sim t_{n - \text{rank } X}.$$

F -test for $H_0 : C\boldsymbol{\beta} = d$, $\text{rank } C = q$

$$F = \frac{(C\hat{\boldsymbol{\beta}} - d)'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\boldsymbol{\beta}} - d)/q}{\hat{\sigma}^2} \sim F_{q, n - \text{rank } X}.$$

Full vs. reduced model

Partition $X = (X_A, X_B)$ and test $H_0 : \boldsymbol{\beta}_B = 0$. Let

$$\text{SSR}_r = \mathbf{y}'(I - P_{X_A})\mathbf{y}, \quad \text{SSR}_{ur} = \mathbf{y}'(I - P_X)\mathbf{y}, \quad q = \text{rank}(X) - \text{rank}(X_A).$$

Then

$$F = \frac{(\text{SSR}_r - \text{SSR}_{ur})/q}{\text{SSR}_{ur}/(n - \text{rank } X)}.$$

Since $P_X = P_{X_A} + P_{M_{X_A}X_B}$, $M_{X_A} = I - P_{X_A}$, the numerator is $\mathbf{y}'(P_X - P_{X_A})\mathbf{y}$ and is independent of $\mathbf{y}'(I - P_X)\mathbf{y}$.

Alternative view: full model vs. reduced model

Consider the partitioned normal model

$$\mathbf{y} = X_A\boldsymbol{\beta}_A + X_B\boldsymbol{\beta}_B + \boldsymbol{\epsilon}, \quad X = (X_A, X_B), \quad H_0 : \boldsymbol{\beta}_B = 0.$$

The two competing mean spaces are

$$\text{restricted: } \mathcal{C}(X_A), \quad \text{unrestricted: } \mathcal{C}(X).$$

With $P_A = P_{X_A}$, $M_A = I - P_A$, and $R_B = M_AX_B$,

$$\mathcal{C}(X) = \mathcal{C}(X_A) \oplus \mathcal{C}(R_B), \quad P_X = P_A + P_{R_B},$$

so the extra sum of squares is exactly

$$\text{SSR}_r - \text{SSR}_{ur} = \mathbf{y}'(P_X - P_A)\mathbf{y} = \mathbf{y}'P_{R_B}\mathbf{y}, \quad q = \text{rank}(R_B) = \text{rank}(X) - \text{rank}(X_A).$$

Under H_0 ,

$$\frac{\mathbf{y}'P_{R_B}\mathbf{y}}{\sigma^2} \sim \chi_q^2, \quad \frac{\mathbf{y}'(I - P_X)\mathbf{y}}{\sigma^2} \sim \chi_{n - \text{rank}(X)}^2, \quad P_{R_B}(I - P_X) = 0,$$

hence the two quadratic forms are independent and the partial- F statistic above follows.

For a general linear hypothesis $H_0 : C\boldsymbol{\beta} = d$, take C full row rank and choose U with orthonormal columns spanning $\ker(C)$. Since

$$I = P_{C'} + UU', \quad P_{C'} = C'(CC')^{-1}C,$$

write

$$\begin{aligned} \tilde{\mathbf{y}} &= \mathbf{y} - XC'(CC')^{-1}d \\ &= \underbrace{XC'(CC')^{-1}}_{Z_A} \underbrace{(C\boldsymbol{\beta} - d)}_{\gamma_A} + \underbrace{XU}_{Z_B} \underbrace{U'\boldsymbol{\beta} + \boldsymbol{\epsilon}}_{\gamma_B}. \end{aligned}$$

Then $H_0 : C\boldsymbol{\beta} = d$ is equivalent to $H_0 : \gamma_A = 0$. Apply the same full-vs-reduced test to the block model with nuisance design Z_B and tested design Z_A :

$$\text{SSE}_r = \tilde{\mathbf{y}}'(I - P_{Z_B})\tilde{\mathbf{y}}, \quad \text{SSE}_{ur} = \tilde{\mathbf{y}}'(I - P_{[Z_B, Z_A]})\tilde{\mathbf{y}}.$$

The numerator degrees of freedom are $q_r = \text{rank}([Z_B, Z_A]) - \text{rank}(Z_B)$, which equals the number of independent estimable constraints, usually $q = \text{rank}(C)$. Equivalently, $X(I - P_{C'})$ may be used for Z_B after dropping dependent columns.

Small-sample alternatives: bootstrap and permutation

When n is small, build the null distribution of a statistic T directly; T may be a t -, Wald-, or partial- F statistic.

Residual bootstrap for ordinary block test $H_0 : \boldsymbol{\beta}_B = 0$. Fit the restricted model $\mathbf{y} = X_A\boldsymbol{\beta}_A + \boldsymbol{\epsilon}$ and compute centered residuals $r_i^0 = y_i - x_{Ai}'\hat{\boldsymbol{\beta}}_A^0$. For $b = 1, \dots, B$:

$$r_1^{*(b)}, \dots, r_n^{*(b)} \sim \text{sample with replacement from } \{r_i^0 - \bar{r}^0\}, \quad y_i^{*(b)} = x_{Ai}'\hat{\boldsymbol{\beta}}_A^0 + r_i^{*(b)}.$$

Refit restricted and unrestricted models to $\mathbf{y}^{*(b)}$ and compute $T^{*(b)}$. For a right-tail F test,

$$p_{\text{boot}} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{T^{*(b)} \geq T_{\text{obs}}\}}{B + 1}.$$

Use a wild bootstrap instead if heteroskedasticity is the main concern.

Permutation test. If $H_0 : \boldsymbol{\beta} = 0$ has no nuisance regressors and errors are exchangeable, permute the response: $\mathbf{y}^* = \Pi\mathbf{y}$, refit, and compare T^* to T_{obs} . With nuisance X_A , use residual permutation / Freedman–Lane:

$$r^0 = M_A\mathbf{y}, \quad \mathbf{y}^* = X_A\hat{\boldsymbol{\beta}}_A^0 + \Pi r^0,$$

then compute the same partial- F statistic from $\mathbf{y}^*$. The permutation p -value has the same form as above, with permutations replacing bootstrap samples. Exactness requires exchangeability under H_0 ; otherwise it is a finite-sample approximation.

Extension to $H_0 : C\boldsymbol{\beta} = d$. First form the transformed block model

$$\tilde{\mathbf{y}} = Z_A\gamma_A + Z_B\gamma_B + \boldsymbol{\epsilon}, \quad H_0 : \gamma_A = 0,$$

as in the previous subsection. Then apply the ordinary procedure with

$$\text{nuisance } X_A \leftarrow Z_B, \quad \text{tested block } X_B \leftarrow Z_A, \quad \text{response } \mathbf{y} \leftarrow \tilde{\mathbf{y}}.$$

For bootstrap, resample centered residuals from the restricted fit $\tilde{\mathbf{y}} \sim Z_B$; for permutation, permute $M_{Z_B}\tilde{\mathbf{y}}$ and add back $P_{Z_B}\tilde{\mathbf{y}}$.

4. Fixed-effects two-factor models

Let A have a levels and B have b levels. Observations are y_{ijk} , with n_{ij} replications in cell (i, j) .

Cell-means and effects parameterizations

Cell-means model:

$$y_{ijk} = \mu_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2).$$

If $n_{ij} \geq 1$,

$$\hat{\mu}_{ij} = \bar{y}_{ij}, \quad \text{Var}(\hat{\mu}_{ij}) = \sigma^2/n_{ij}, \quad \widehat{\sum_{ij} c_{ij}\mu_{ij}} = \sum_{ij} c_{ij}\bar{y}_{ij}.$$

Effects parameterization:

$$\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij},$$

The design matrix is rank-deficient, but estimable functions of cell means are unchanged. Important contrasts:

$$\text{simple } A \text{ effect at } B = j : \mu_{ij} - \mu_{i'j},$$

$$A \text{ main effect : } \bar{\mu}_{i\cdot} - \bar{\mu}_{i'\cdot}, \quad \bar{\mu}_{i\cdot} = b^{-1} \sum_j \mu_{ij},$$

$$AB \text{ interaction : } \mu_{ij} - \mu_{i'j} - \mu_{i'j'} + \mu_{ij'}.$$

LSMEANS use equal-weight marginal averaging; they equal raw marginal means only under balance.

Additive model

If $\gamma_{ij} = 0$,

$$y_{ijk} = \mu + \alpha_i + \beta_j + \varepsilon_{ijk},$$

so simple effects equal main effects. With missing cells, $\mu + \alpha_i + \beta_j$ remains estimable if the two-factor incidence graph is connected.

Sequential ANOVA and type I/II/III SS

Let

$$X_1 = \mathbf{1}_n, \quad X_2 = [\mathbf{1}_n, A], \quad X_3 = [X_2, B], \quad X_4 = X_{AB} \quad (\text{cell-means matrix}),$$

with $\mathcal{C}(X_i) \subset \mathcal{C}(X_2) \subset \mathcal{C}(X_3) \subset \mathcal{C}(X_4)$. Put $P_j = P_{X_j}$.

Source	Sequential SS	df
$A \mid 1$	$\mathbf{y}'(P_2 - P_1)\mathbf{y}$	$a - 1$
$B \mid 1, A$	$\mathbf{y}'(P_3 - P_2)\mathbf{y}$	$b - 1$
$AB \mid 1, A, B$	$\mathbf{y}'(P_4 - P_3)\mathbf{y}$	$(a - 1)(b - 1)$
Error	$\mathbf{y}'(I - P_4)\mathbf{y}$	$n - ab$
Corrected total	$\mathbf{y}'(I - P_1)\mathbf{y}$	$n - 1$

- Balanced cells: increments $\mathcal{C}(X_{j+1}) \ominus \mathcal{C}(X_j)$ are orthogonal, so type I, II, and III SS coincide and test the usual main effects/interactions.
- Unbalanced cells: type I depends on order; type II tests each term after all same/lower-order terms; type III tests each term after all other terms and corresponds to LSMEAN contrasts.
- The highest-order interaction test is the same for type I/II/III.

Type I=II=III in balanced designs

With $n_{ij} \equiv n$ replicates per cell, the Kronecker basis (Sec. 6) yields orthogonal blocks $X_S'X_{S'} = 0$ for $S \neq S'$, hence

$$P_X = P_1 + P_A + P_B + P_{AB},$$

so $\text{SS}_S = \mathbf{y}'P_S\mathbf{y}$ is invariant to fitting order. Direct check for A : B -only fit $\hat{y}_{ijk}^{(B)} = \bar{y}_{j\cdot}$, additive fit $\hat{y}_{ijk}^{(A,B)} = \bar{y}_{i\cdot} + \bar{y}_{j\cdot} - \bar{y}_{\dots}$, full fit $\hat{y}_{ijk}^{(F)} = \bar{y}_{ij\cdot}$, and the (B, AB) -restricted fit $\hat{y}_{ijk}^{(B,AB)} = \bar{y}_{ij\cdot} - \bar{y}_{i\cdot} + \bar{y}_{\dots}$. Each increment equals $\bar{y}_{i\cdot} - \bar{y}_{\dots}$, independent of j, k , giving

$$\text{SS}_{A|1} = \text{SS}_{A|1,B} = \text{SS}_{A|1,B,AB} = nb \sum_i (\bar{y}_{i\cdot} - \bar{y}_{\dots})^2.$$

Type III in unbalanced designs: LSMEAN contrasts

Cell-means form $\mathbf{y} = Z\boldsymbol{\mu} + \boldsymbol{\epsilon}$ with $\boldsymbol{\mu} \in \mathbb{R}^{ab}$ and $(Z'Z)^{-1}Z'\mathbf{y} = (\bar{y}_{ij\cdot})$. Type III imposes $C\boldsymbol{\mu} = 0$. Lagrangian constrained LS gives

$$\hat{\boldsymbol{\mu}}_c = (Z'Z)^{-1}Z'\mathbf{y} - (Z'Z)^{-1}C'\{C(Z'Z)^{-1}C'\}^{-1}C(Z'Z)^{-1}Z'\mathbf{y},$$

and the partial- F projection

$$P_F - P_R = Z(Z'Z)^{-1}C'\{C(Z'Z)^{-1}C'\}^{-1}C(Z'Z)^{-1}Z'.$$

With first-difference D_a and centering $M_a = I_a - a^{-1}\mathbf{1}_a\mathbf{1}_a'$,

$$C_A = D_a(I_a \otimes b^{-1}\mathbf{1}_b'), \quad C_{AB} = M_a \otimes M_b,$$

so $C_A\boldsymbol{\mu} = 0$ tests equality of equal-weight marginals $b^{-1} \sum_j \mu_{ij}$, and $(C_{AB}M_b)_{ij} = \mu_{ij} - \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot j} + \bar{\mu}_{\dots}$ is the interaction contrast. Because C acts on cells with weights $1/b$, $1/a$ rather than n_{ij} -proportional, Type III F -tests are LSMEAN contrasts and probe the underlying μ_{ij} unaffected by cell imbalance.

5. Partial correlation

Let $X = (X_1, \dots, X_p)'$ and partition $A = (X_1, X_2)'$, $Z = (X_3, \dots, X_p)'$. The partial correlation of X_1 and X_2 given Z has equivalent forms:

$$\begin{aligned}\rho_{12|Z} &= -\frac{\omega_{12}}{\sqrt{\omega_{11}\omega_{22}}}, & \Omega &= \Sigma^{-1} = (\omega_{ij}), \\ \rho_{12|Z} &= -\frac{s_{12}}{\sqrt{s_{11}s_{22}}}, & S &= \Sigma_{AA} - \Sigma_{AZ}\Sigma_{ZZ}^{-1}\Sigma_{ZA}, \\ \rho_{12|Z} &= \text{Corr}(R_1, R_2), & R_i &= X_i - \text{Proj}_{\text{span}(Z)} X_i.\end{aligned}$$

For $p = 3$,

$$\rho_{12|3} = \frac{\rho_{12} - \rho_{13}\rho_{23}}{\sqrt{(1 - \rho_{13}^2)(1 - \rho_{23}^2)}}.$$

6. Balanced factorials via Kronecker products

For balanced n -factor designs with a_i levels and r replications per cell, choose a contrast basis $C_{a_i} \in \mathbb{R}^{a_i \times (a_i-1)}$ satisfying $\mathbf{1}'_{a_i} C_{a_i} = 0$ and $\text{rank } C_{a_i} = a_i - 1$. For an effect indexed by $S \subseteq \{1, \dots, n\}$,

$$X_S = \left(\bigotimes_{i \in S} Z_i^{(S)}\right) \otimes \mathbf{1}_r, \quad Z_i^{(S)} = \begin{cases} C_{a_i}, & i \in S, \\ \mathbf{1}_{a_i}, & i \notin S. \end{cases}$$

Thus $S = \emptyset$ is the grand mean, $|S| = 1$ gives main effects, $|S| = 2$ gives two-way interactions, etc. The block dimension is

$$p_S = \prod_{i \in S} (a_i - 1), \quad \sum_S p_S = \prod_i a_i.$$

Main consequence. If $S \neq S'$, then $X'_S X_{S'} = 0$ because at least one Kronecker factor is $C'_{a_i} \mathbf{1}_{a_i} = 0$. Hence

$$X'X = \bigoplus_S X'_S X_S, \quad \hat{\theta}_S = (X'_S X_S)^{-1} X'_S \mathbf{y}, \quad \text{SS}_S = \mathbf{y}' X_S (X'_S X_S)^{-1} X'_S \mathbf{y}.$$

Type I, II, and III SS coincide in balanced orthogonal factorials. The pure error term is

$$\text{SSE} = \mathbf{y}'(I_{\prod_i a_i} \otimes Q_r) \mathbf{y}, \quad Q_r = I_r - r^{-1} \mathbf{1}_r \mathbf{1}'_r, \quad \text{df}_E = (r - 1) \prod_i a_i.$$

Useful identities: $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ and $(A \otimes B)' = A' \otimes B'$.

7. Linear mixed-effects models Framework

$$\mathbf{y} = X\boldsymbol{\beta} + Z\mathbf{u} + \boldsymbol{\varepsilon}, \quad \begin{pmatrix} \mathbf{u} \\ \boldsymbol{\varepsilon} \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} G & 0 \\ 0 & R \end{pmatrix}\right).$$

Then

$$\mathbb{E}\mathbf{y} = X\boldsymbol{\beta}, \quad \Sigma \equiv \text{Var}(\mathbf{y}) = ZGZ' + R, \quad \mathbb{E}(\mathbf{y} \mid \mathbf{u}) = X\boldsymbol{\beta} + Z\mathbf{u}, \quad \text{Var}(\mathbf{y} \mid \mathbf{u}) = R.$$

With independent random factors $Z = [Z_1, \dots, Z_m]$, $\text{Var}(\mathbf{u}_j) = \sigma_j^2 I$, and $R = \sigma_e^2 I$,

$$\text{Var}(\mathbf{y}) = \sum_{j=1}^m \sigma_j^2 Z_j Z'_j + \sigma_e^2 I.$$

GLS vs. OLS

If $\Sigma = \sigma^2 V$ and V is known,

$$\widehat{C}\widehat{\boldsymbol{\beta}}_V = C(X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y} \quad (\text{BLUE}).$$

In balanced orthogonal designs, the GLS and OLS estimators of estimable fixed effects coincide. In unbalanced designs, BLUEs become precision-weighted averages and generally differ from raw means.

Asymptotics for GLS in nested subsampling

For the nested subsampling model

$$y_{ijk} = x'_{ij} \theta + u_{ij} + \varepsilon_{ijk}, \quad i = 1, \dots, t, \quad j = 1, \dots, n_i, \quad k = 1, \dots, m_{ij},$$

assume independent u_{ij} and ε_{ijk} with $\mathbb{E}u_{ij} = \mathbb{E}\varepsilon_{ijk} = 0$, $\text{Var}(u_{ij}) = \sigma_u^2$, $\text{Var}(\varepsilon_{ijk}) = \sigma_e^2$. Averaging within experimental unit gives

$$\bar{y}_{ij.} = x'_{ij} \theta + \eta_{ij}, \quad \eta_{ij} = u_{ij} + \bar{\varepsilon}_{ij.}, \quad v_{ij} = \text{Var}(\eta_{ij}) = \sigma_u^2 + \frac{\sigma_e^2}{m_{ij}}.$$

Let $N = \sum_i n_i$, $\bar{\mathbf{y}} = (\bar{y}_{ij.})$, $\bar{X} = (x'_{ij.})$, and $V = \text{diag}(v_{ij.})$. The oracle GLS estimator is

$$\hat{\theta}_G = (\bar{X}'V^{-1}\bar{X})^{-1}\bar{X}'V^{-1}\bar{\mathbf{y}} = \theta + (\bar{X}'V^{-1}\bar{X})^{-1}\bar{X}'V^{-1}\eta,$$

equivalently minimizing $\sum_{ij} v_{ij}^{-1}(\bar{y}_{ij.} - x'_{ij} \theta)^2$. A feasible version uses

$$\hat{\theta}_{OLS} = (\bar{X}'\bar{X})^{-1}\bar{X}'\bar{\mathbf{y}}, \quad \hat{\eta}_{ij} = \bar{y}_{ij.} - x'_{ij} \hat{\theta}_{OLS},$$

and estimates variance components by the moment regression

$$(\hat{\sigma}_u^2, \hat{\sigma}_e^2) = \arg \min_{a,b} \sum_{ij} \left(\hat{\eta}_{ij}^2 - a - \frac{b}{m_{ij}} \right)^2, \quad \hat{V} = \text{diag} \left(\hat{\sigma}_u^2 + \frac{\hat{\sigma}_e^2}{m_{ij}} \right).$$

Then replace V by $\hat{V}$ in GLS.

Regime 1: many experimental units, bounded subsamples. If each $n_i \rightarrow \infty$, $m_{ij} \leq C$, and

$$\frac{1}{N} \bar{X}'V^{-1}\bar{X} \rightarrow Q_V \succ 0, \quad \max_{ij} \frac{\|x_{ij}\|}{\sqrt{N}} \rightarrow 0,$$

then

$$\sqrt{N}(\hat{\theta}_G - \theta) = \left(\frac{1}{N} \bar{X}'V^{-1}\bar{X} \right)^{-1} \frac{1}{\sqrt{N}} \bar{X}'V^{-1}\eta \xrightarrow{d} \mathcal{N}(0, Q_V^{-1}).$$

If $\hat{\sigma}_u^2 \xrightarrow{p} \sigma_u^2$ and $\hat{\sigma}_e^2 \xrightarrow{p} \sigma_e^2$, then plug-in weights do not affect the first-order limit:

$$\frac{1}{\sqrt{N}} \bar{X}'\hat{V}^{-1}\eta = \frac{1}{\sqrt{N}} \bar{X}'V^{-1}\eta + \frac{1}{\sqrt{N}} \sum_{ij} x_{ij} \eta_{ij} \left(\hat{v}_{ij}^{-1} - v_{ij}^{-1} \right),$$

where the second term is $o_p(1)$ under uniform consistency of $\hat{v}_{ij}^{-1}$. Hence

$$\sqrt{N}(\hat{\theta}_{FGLS} - \hat{\theta}_G) = o_p(1), \quad \sqrt{N}(\hat{\theta}_{FGLS} - \theta) \xrightarrow{d} \mathcal{N}(0, Q_V^{-1}).$$

Thus the effective sample size is the number of independent experimental units N , not the number of subsamples $\sum_{ij} m_{ij}$.

Regime 2: bounded experimental units, many subsamples. If $n_i \leq C$ while $m_{ij} \rightarrow \infty$, then $\varepsilon_{ij.} \xrightarrow{p} 0$ and

$$\bar{y}_{ij.} \xrightarrow{p} x'_{ij} \theta + u_{ij}, \quad v_{ij} \rightarrow \sigma_u^2.$$

With only finitely many independent u_{ij} 's, increasing subsamples estimates the unit means $x'_{ij} \theta + u_{ij}$, not the population fixed effect $x'_{ij} \theta$ precisely. If N is fixed and $\bar{X}$ has full column rank, then conditionally on $\mathbf{u}$,

$$\hat{\theta}_G \xrightarrow{p} \theta + (\bar{X}'\bar{X})^{-1}\bar{X}'\mathbf{u},$$

which is a random limit. If $\mathbf{u} \sim \mathcal{N}(0, \sigma_u^2 I)$, this limit has variance $\sigma_u^2 (\bar{X}'\bar{X})^{-1}$. For treatment means, for example,

$$\text{Var}(\bar{y}_{i..}) = \frac{1}{n_i^2} \sum_{j=1}^{n_i} \left(\sigma_u^2 + \frac{\sigma_e^2}{m_{ij}} \right) \rightarrow \frac{\sigma_u^2}{n_i}.$$

Therefore $m_{ij} \rightarrow \infty$ alone does not give consistency for fixed effects unless $\sigma_u^2 = 0$ or the number of independent experimental units also diverges.

ANOVA and EMS recipe

For nested model spaces $\mathcal{C}(X_1) \subset \dots \subset \mathcal{C}(X_K)$,

$$\text{SS}_{j+1|j} = \mathbf{y}'(P_{j+1} - P_j)\mathbf{y}, \quad \text{df}_{j+1|j} = \text{rank}(X_{j+1}) - \text{rank}(X_j).$$

Expected mean squares can be computed from

$$\mathbb{E}(\mathbf{y}'A\mathbf{y}) = \text{tr}(A\Sigma) + (\mathbb{E}\mathbf{y})'(A\mathbb{E}\mathbf{y}).$$

For iid $w_1, \dots, w_k$ with variance σ_w^2 , $\mathbb{E}\sum_i (w_i - \bar{w})^2 = (k - 1)\sigma_w^2$.

8. ANOVA templates for common randomization structures

In the tables below, A, B, C are fixed treatment factors. Random terms are independent Gaussian errors. For a fixed source T , write Λ_T for its fixed-effect noncentral contribution; under $H_0(T)$, $\Lambda_T = 0$.

8.1 CRD with subsampling

Design. a treatments, n experimental units per treatment, m subsamples per experimental unit:

$$y_{ijk} = \mu + \alpha_i + u_{ij} + e_{ijk}, \quad i = 1, \dots, a, \quad j = 1, \dots, n, \quad k = 1, \dots, m,$$

$$u_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_u^2), \quad e_{ijk} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_e^2), \quad u \perp\!\!\!\perp e.$$

Source	df	SS shortcut	EMS	Test / denominator
A	$a - 1$	$nm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2$	$\sigma^2 + m\sigma^2 + \Lambda_A/(a - 1)$	$\text{MS}_A/\text{MS}_{U(A)}$
$U(A)$: units within treatment	$a(n - 1)$	$m \sum_{ij} (\bar{y}_{ij.} - \bar{y}_{i..})^2$	$\sigma_e^2 + m\sigma_u^2$	error for A
Subsampling error	$an(m - 1)$	$\sum_{ijk} (y_{ijk} - \bar{y}_{ij.})^2$	σ_e^2	—
Corrected total	$anm - 1$	$\sum_{ijk} (y_{ijk} - \bar{y}_{...})^2$	—	—

Variance components and treatment contrasts:

$$\hat{\sigma}_e^2 = \text{MS}_E, \quad \hat{\sigma}_u^2 = \frac{\text{MS}_{U(A)} - \text{MS}_E}{m}.$$

For $L = \sum_i l_i \alpha_i$ with $\sum_i l_i = 0$,

$$\hat{L} = \sum_i l_i \bar{y}_{i..}, \quad \widehat{\text{Var}}(\hat{L}) = \frac{\text{MS}_{U(A)}}{nm} \sum_i l_i^2, \quad \text{df} = a(n - 1).$$

8.2 Split-plot design

Design. r blocks, whole-plot factor A with a levels, split-plot factor B with b levels. One whole plot for each (block, A); all B levels appear inside each whole plot:

$$y_{ijk} = \mu + \rho_k + \alpha_i + w_{ik} + \beta_j + (\alpha\beta)_{ij} + e_{ijk},$$

$$i = 1, \dots, a, \quad j = 1, \dots, b, \quad k = 1, \dots, r, \quad w_{ik} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_w^2), \quad e_{ijk} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_e^2).$$

Here ρ_k is a block nuisance effect; if blocks are random, only the block row EMS changes.

Source	df	SS shortcut	EMS	Test / denominator
Block	$r - 1$	$ab \sum_k (\bar{y}_{..k} - \bar{y}_{...})^2$	nuisance	—
A whole-plot	$a - 1$	$b \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2$	$\sigma^2 + b\sigma^2 + \Lambda_A/(a - 1)$	$\text{MS}_A/\text{MS}_{\text{Blok} \times A}$
Block $\times A$ whole-plot error	$(r - 1)(a - 1)$	$b \sum_{ik} (\bar{y}_{i.k} - \bar{y}_{i..})^2$	$\sigma_e^2 + b\sigma_w^2$	error for A
B split-plot	$b - 1$	$a \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2$	$\sigma^2 + \Lambda_B/(b - 1)$	MS_B/MS_E
$A \times B$	$(a - 1)(b - 1)$	$r \sum_{ij} (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2$	$\sigma^2 + \Lambda_{AB}/[(a - 1)(b - 1)]$	$\text{MS}_{AB}/\text{MS}_E$
Split-plot error	$a(r - 1)(b - 1)$	$\sum_{ijk} (y_{ijk} - \bar{y}_{i.k} - \bar{y}_{ij.} + \bar{y}_{i..} + \bar{y}_{.j.} - \bar{y}_{...})^2$	σ_e^2	—
Corrected total	$abr - 1$	$\sum_{ijk} (y_{ijk} - \bar{y}_{...})^2$	—	—

Useful standard errors for contrasts:

Estimand	Estimated variance	df
$\sum_i l_i \bar{\mu}_i, \sum_i l_i = 0$	$\frac{\text{MS}_{\text{Blok} \times A}}{br} \sum_i l_i^2$	$(r - 1)(a - 1)$
$\sum_j m_j \bar{\mu}_{.j}, \sum_j m_j = 0$	$\frac{br}{\text{MS}_E} \sum_j m_j^2$	$a(r - 1)(b - 1)$
B contrast within fixed $A = i$	$\frac{br}{\text{MS}_E} \sum_j m_j^2$	$a(r - 1)(b - 1)$
A contrast within fixed $B = j$	$\frac{1}{r} \frac{\text{MS}_{\text{Blok} \times A} + \frac{b-1}{b} \text{MS}_E}{r} \sum_i l_i^2$	C- S .

8.3 Split-split-plot design

Design. r blocks; whole-plot factor A with a levels; split-plot factor B with b levels; split-split-plot factor C with c levels:

$$y_{ijk\ell} = \mu + \rho_\ell + \alpha_i + u_{i\ell} + \beta_j + (\alpha\beta)_{ij} + v_{ij\ell} + \gamma_k + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + (\alpha\beta\gamma)_{ijk} + e_{ijk\ell},$$

where $i = 1, \dots, a$, $j = 1, \dots, b$, $k = 1, \dots, c$, $\ell = 1, \dots, r$, and

$$u_{i\ell} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_u^2), \quad v_{ij\ell} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_v^2), \quad e_{ijk\ell} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_e^2).$$

Source	df	EMS	Test / denominator
Block	$r - 1$	nuisance	—
A whole-plot	$a - 1$	$\sigma^2 + c\sigma^2 + bc\sigma^2 + \Lambda_A/(a - 1)$	$\text{MS}_A/\text{MS}_{\text{Blok} \times A}$
Block $\times A$ whole-plot error	$(r - 1)(a - 1)$	$\sigma^2 + c\sigma^2 + bc\sigma^2$	error for A
B split-plot	$b - 1$	$\sigma^2 + c\sigma^2 + \Lambda_B/(b - 1)$	$\text{MS}_B/\text{MS}_{\text{Blok} \times A \times B}$
$A \times B$	$(a - 1)(b - 1)$	$\sigma^2 + c\sigma^2 + \Lambda_{AB}/[(a - 1)(b - 1)]$	$\text{MS}_{AB}/\text{MS}_{\text{Blok} \times A \times B}$
Block $\times A \times B$ split-plot error	$a(r - 1)(b - 1)$	$\sigma_e^2 + c\sigma_v^2$	error for B, AB
C split-split-plot	c	$\sigma_v^2 + \Lambda_C/(c - 1)$	MS_C/MS_E
$A \times C$	$(a - 1)(c - 1)$	$\sigma^2 + \Lambda_{AC}/[(a - 1)(c - 1)]$	$\text{MS}_{AC}/\text{MS}_E$
$B \times C$	$(b - 1)(c - 1)$	$\sigma^2 + \Lambda_{BC}/[(b - 1)(c - 1)]$	$\text{MS}_{BC}/\text{MS}_E$
$A \times B \times C$	$(a - 1)(b - 1)(c - 1)$	$\sigma^2 + \Lambda_{ABC}/[(a - 1)(b - 1)(c - 1)]$	$\text{MS}_{ABC}/\text{MS}_E$
Split-split error	$ab(r - 1)(c - 1)$	σ_e^2	—
Corrected total	$abcr - 1$	—	—

Compact SS shortcuts:

$$\begin{aligned}\text{SS}_A &= bcr \sum (\bar{y}_{i...} - \bar{y}_{...})^2, \\ \text{SS}_{\text{Blok} \times A} &= bc \sum_{i\ell} (\bar{y}_{i..\ell} - \bar{y}_{i...} - \bar{y}_{..\ell} + \bar{y}_{...})^2, \\ \text{SS}_B &= acr \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2, \\ \text{SS}_{AB} &= cr \sum_{ij} (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2, \\ \text{SS}_C &= abr \sum_k (\bar{y}_{..k} - \bar{y}_{...})^2,\end{aligned}$$

with analogous lower-order-removal formulas for AC , BC , and ABC . Error SS are the corresponding residual sums of squares in their randomization strata.

Variance-component estimators:

$$\hat{\sigma}_e^2 = \text{MS}_E, \quad \hat{\sigma}_v^2 = \frac{\text{MS}_{\text{Blok} \times A \times B} - \text{MS}_E}{c}, \quad \hat{\sigma}_u^2 = \frac{\text{MS}_{\text{Blok} \times A} - \text{MS}_{\text{Blok} \times A \times B}}{bc}.$$

9. Cochran–Satterthwaite approximation

For an independent linear combination $\hat{\theta} = \sum_\ell c_\ell \text{MS}_\ell$, with $\text{df}_\ell \text{MS}_\ell / \mathbb{E}(\text{MS}_\ell) \sim \chi^2_{\text{df}_\ell}$,

$$\widehat{\text{df}} = \frac{(\sum_\ell c_\ell \text{MS}_\ell)^2}{\sum_\ell (c_\ell^2 \text{MS}_\ell) / \text{df}_\ell}$$

and $\hat{\theta} \widehat{\text{df}} / \theta \sim \chi^2_{\widehat{\text{df}}}$. Use this whenever the denominator for a t or F statistic is a linear combination of mean squares.

Approximate F recipe. Choose $\hat{\theta} = \sum_\ell c_\ell \text{MS}_\ell$ so that $\mathbb{E}(\hat{\theta})$ matches the nuisance variance part of the numerator EMS under H_0 . Then

$$\frac{\text{MS}_{\text{num}}}{\hat{\theta}} \sim F_{\text{df}_{\text{num}}, \widehat{\text{df}}}.$$